

ABHINAV GOEL

Robotics Engineer | Autonomous Systems | UAV Engineering

+91-8384857368 | abhinav.goel.robotics@gmail.com | linkedin.com/in/abhinav-g-1b878a2b5

github.com/Humobot1812 | Muzaffarnagar, Uttar Pradesh, India

AG

SUMMARY

Robotics Engineer specializing in ROS2-based autonomous navigation, SLAM, and vision-based UAV systems, with a proven record of building robots from simulation to field-tested hardware. Skilled in sensor fusion (LiDAR, IMU, encoders), real-time computer vision (OpenCV, YOLOv8), and drone autopilot integration (ArduPilot, MAVSDK, MAVLink). **Top 16 / 400+ teams** (top 4% nationally) at IIT Bombay e-Yantra 2025–26. Seeking an internship or research role in robotics or UAV engineering.

EXPERIENCE

Coordinator — Electronics and Robotics Society

PDPM IITDM Jabalpur

08/2024 – Present | Jabalpur

- Led and scaled the society's technical program to **200+ active members**, organizing 10+ robotics workshops and lab sessions per semester.
- Mentored **30+ junior students** in ROS2, embedded systems, and robotics, accelerating their readiness for national competitions.
- Coordinated multi-team participation in e-Yantra at IIT Bombay, contributing to a **Top 16 national ranking**.

Core Committee Member — Aero Fabrication Club

PDPM IITDM Jabalpur

08/2024 – Present | Jabalpur

- Designed and fabricated UAV hardware (frame assembly, ESC wiring, flight controller integration) for competition-ready drones.
- Represented the club in **2 national UAV competitions** (SAE Aerothon 2025, NIDAR 2026), contributing to autonomous mission design and hardware testing.

EDUCATION

B.Tech — Electronics and Communication Engineering

PDPM IITDM Jabalpur

08/2024 – 08/2028 | Jabalpur

Coursework: Robotics (kinematics, D-H params, trajectory planning), Sensors & Actuators, Control Systems, Embedded Systems. Lab: PUMA robot kinematics; trajectory-following mobile robot.

PROJECTS

e-Yantra KrishiCobot — IIT Bombay (e-YRC 2025–26, Stage 2)

08/2025 – 12/2025 | IIT Bombay

ROS2 Humble, LiDAR, IMU, Encoders, OpenCV, ArUco Markers, UR5 Arm, Visual Servoing

- Engineered a fully autonomous agricultural robot for crop monitoring; achieved **Top 16 / 400+ teams** (top 4%) at IIT Bombay.
- Built dual-sensor (LiDAR + ultrasonic) perception pipeline enabling collision-free navigation across all field lanes.
- Developed ArUco-based localization to classify healthy vs. defective fruits; automated pick-and-place via UR5 arm with visual servoing.

Disaster Surveillance Drone — NIDAR 2026

01/2026 – 06/2026 | National UAV Competition

Python, MAVSDK, ArduPilot SITL, OpenCV, YOLOv8, Shapely, KML

- Designed a polygon-based survey system covering **75 acres in <20 min** at 25 m altitude via optimized zigzag planning.
- Integrated YOLOv8 real-time human detection with RTL failsafe and pause-center-resume logic; zero missed detections.
- Fully automated mission lifecycle via MAVSDK API, reducing operator workload to zero during flight.

Autonomous Drone Survey System — SAE Aerothon 2025

01/2025 – 06/2025 | National Competition

Python, ROS2, DroneKit, ArduPilot SITL, MAVLink, Gazebo, YOLOv4-Tiny

- Built ROS2-based autonomous drone for survey + real-time object detection; selected for national-level presentation.
- Implemented pinhole camera pixel-to-GPS conversion enabling **sub-meter target geolocation** from aerial imagery.
- Engineered PID-based visual centering loop for precise autonomous target alignment during hover.

Differential Drive Robot — ROS2 Navigation and SLAM

01/2024 – 06/2024

ROS2 Humble, Gazebo, Nav2, SLAM Toolbox, AMCL, URDF, TF2, EKF

- Deployed complete autonomous nav stack (Nav2 + EKF) with reliable obstacle avoidance in simulated environments.
- Achieved SLAM-based 2D mapping + AMCL localization enabling GPS-free navigation in unknown environments.
- Produced YouTube tutorial series on full pipeline; earned **6 GitHub Stars**.

ESP32-Based Quadcopter Drone — Elective Project

06/2024 – 12/2024

ESP32-S3, MPU6050 IMU, Barometer, Coreless DC Motors, Custom MOSFET ESC

- Fabricated a quadcopter from scratch with IMU + barometer altitude sensing and a custom MOSFET-based ESC.
- Flashed manual flight control firmware on ESP32-S3 using Embedded C; achieved stable hover in test conditions.



KEY ACHIEVEMENTS

★ e-Yantra Robotics Competition

Top 16 / 400+ teams in e-YRC 2025–26 at IIT Bombay (top 4% nationally).

🏆 SAE Aerothon 2025

Selected to develop and present an autonomous drone survey and target detection system nationally.

■ NIDAR 2026

Disaster Surveillance Drone selected for the national UAV challenge.

SKILLS

Languages

Python · C++ · Embedded C

Robotics

ROS2 Humble · Nav2 · MoveIt2
SLAM Toolbox · AMCL · TF2

Drones

ArduPilot · MAVLink · MAVSDK
DroneKit · PX4

Simulation

Gazebo Harmonic · RViz2

Vision

OpenCV · YOLOv4 · YOLOv8
ArUco Markers · Visual Servoing

Control

EKF · PID · Sensor Fusion
URDF · xacro

Hardware

LiDAR · IMU · ESP32 · Arduino
Custom MOSFET ESC

Tools

Ubuntu 22.04 · Git · NumPy
Matplotlib · Pandas · KML